SMART INDIA HACKATHON 2026 — MASTER TECHNICAL DOSSIER

Team CIPHORA (RMKSIH26SW203) | Problem Statement: SIH26117 | Theme: Smart Automation

1. EXECUTIVE SUMMARY & SOVEREIGNTY GUARANTEE

The Sovereign On-Premise Agentic AI Workbench is an enterprise platform engineered for air-gapped industrial facilities (Refineries, Power PSUs, and Defense Units). All AI processing, OCR extraction, vector embeddings, code sandbox execution, and Microsoft Word/Excel file generation execute 100% locally on-premise (EXTERNAL_AI_ENABLED=false) with zero outbound network calls.

2. TECHNICAL APPROACH & STACK JUSTIFICATION (PPT SLIDE 3 ALIGNMENT)

Component	Technology Stack	Why We Used It & Architectural Advantages
Frontend UI	React 18 + Vite + Tailwind	Sub-second responsive UI, offline client compilation, dynamic step visualizer.
API Gateway	FastAPI (Python 3.13)	Asynchronous IO event loop, strict Pydantic v2 validation, OpenAPI docs.
Model Router	Dynamic Intent Classifier	Routes coding to qwen2.5-coder, SOPs to qwen2.5-instruct, vision to llama3.2.
Local Vector DB	384d Dense Vectors + SQLite	Sub-50ms cosine similarity search, page-level grounded citations (Page 4).
OCR & Ingestion	pypdf + Local Tesseract	Auto-detects scanned pages (<20 chars) and runs 100% offline OCR.
Agentic Engine	Python Sandbox + docx/xlsx	5-step automated workflow, math calculations, real downloadable files.

3. MASTER SIH JUDGE DEFENSE HIGHLIGHTS (TOP QUESTIONS)

Q1: How do you prove no data leaves the premises?
A: Our application enforces EXTERNAL_AI_ENABLED=false. All servers run on localhost (127.0.0.1 loopback), SQLite is stored on disk, and the live Security & Audit Center monitors Outbound Connections: 0 and External AI Calls: 0.

Q2: Why use a Multi-Model Router instead of one huge 70B model?
A: Heavy 70B models require expensive enterprise GPUs. Our router dispatches tasks to specialized 7B models (qwen2.5-coder for math/code, qwen2.5-instruct for SOPs), enabling sub-100ms execution on standard 12GB–24GB VRAM workstation hardware.

Q3: How are deliverables generated instead of chat replies?
A: We integrated python-docx and openpyxl inside deliverable_generator.py to programmatically create physical .docx Emergency Shutdown Approval Notes and .xlsx spreadsheets, complete with Senior Engineer sign-off gates.